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Quentin Rebergue - Graduate Software Engineer - Systems

Profile

Looking for a full-time position in software engineering with a focus on: Platform, Linux, Systems, Infrastructure. MSc in Computer Engineering (EPITA - GISTRE) in Systems / Embedded. Fluent in French and English (TOEIC 915, C1). Interests: watchmaking, music, hiking/running, cooking.

Work Experience

- **Software Engineer Intern - Platform team, ARISTA**, Dublin 02/2026 – 08/2026

  Designed and implemented full non-zero reset value pipeline for ASIC registers across the chip-agnostic & performance-critical Strata middleware between software agents and ASIC hardware, covering SDK parsing and code generation in Python to C++ compile-time optimization and DMA management, achieving zero runtime overhead. Won 1st place at Arista's ASTERIX Hackathon by building an AI agent context optimization tool in 24h (AST-based, PageRank, MCP server) with 70% token reduction on internal codebases, validated with 400+ tests and custom benchmark. Approved for production development by Director of Engineering.

- **Head of Teaching Assistants - YAKA** / **ACU Program, EPITA**, Kremlin-Bicetre 09/2024 – 02/2026

  Led the national team of 67 TAs and supervised 5 intensive coding bootcamps (Java, C++, JS, C/Bash, SQL) and 7 large-scale projects (compiler, HTTP server in C, myMalloc, POSIX shell) for 700+ 3rd-year students across 5 campuses. Maintained high technical standards through systematic 1-1 code reviews, and improved TA-to-student ratio from 1:30 to 1:20.

School Projects

- **My Docker** – Built a lightweight container runtime in C, using Linux namespaces, pivot_root, capabilities dropping, seccomp syscall filtering, and cgroups. Pulls and mounts a root filesystem from a container registry.
- **RFID Driver** – Developed a Linux kernel character device driver for the MFRC522 RFID reader with file_operations callbacks, SPI regmap communication, and a FIFO command interface. Tested via QEMU emulation.
- **Linux System Tracing & Storage** – Used strace fault injection and syscall filtering to reverse-engineer a binary. Wrote a Bash script automating Linux storage stacks: loop devices, LUKS2, LVM, dm-linear, dm-delay, and RAID.
- **GPU-Accelerated Video Analysis System (IRGPU)** – Built a real-time motion detection pipeline in C++/CUDA (reservoir sampling, morphological ops, hysteresis thresholding). Achieved 20x CPU speedup through iterative CUDA optimization profiled with Nsight Compute. Team of 4.

Education

- **MSc Computer Engineering, Low-level & Systems (GISTRE)**, EPITA 2020 – 2026
- **Preparatory Class (PTSI) in Engineering Science**, Lycée Jules Ferry Versailles 2019 – 2020

Technical Expertise

- **Programming & Languages:** Bash, C, C++, CUDA, Java, OOP, PostgreSQL, Python, SQL, Shell.
- **Other** / **DevOps:** Git, Kubernetes, Security, monitoring, optimisation, strace, system analysis, valgrind, perf.
- **Systems:** ASIC interfacing, Bootloader, DMA, Drivers, FPGA, GPU, Kernel, Linux, Multithreading, Virtualization, low-latency, networking, operating systems, profiling, real-time, virtual memory.